Abinaya K

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Professional Summary

Computer Science Engineering graduate with hands-on experience in Python, Generative AI, LLM applications, Prompt Engineering, and Retrieval-Augmented Generation (RAG). Built AI projects involving document question answering, semantic search, embeddings, LLM API integration, computer vision, and automation. Strong foundation in Machine Learning, Deep Learning, NLP, SQL, and REST APIs, with a keen interest in building practical AI solutions and developing expertise in enterprise applications.

Education

B.E. Computer Science and Engineering Nov 2022 – May 2026
University College of Engineering, BIT Campus, Anna University, Tiruchirappalli

- CGPA: 8.32

Higher Secondary June 2021 – May 2022
Ideal Matric Hr. Sec. School, Aranthangi

- Percentage: 93.67%

Experience

Data Analytics Intern – NUX Software Solutions July – August 2025

- Gained practical experience in data preparation, analysis, and interpretation of structured datasets while working under senior analysts to understand data-driven problem-solving and business decision-making.

Projects

Multimodal GenAI Assistant

- Developed a Python-based Generative AI application integrating Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), semantic search, text-to-image generation, and YouTube video summarization.
- Implemented PDF question answering using text extraction, document chunking, Sentence Transformer embeddings, and FAISS vector search to retrieve relevant context for LLM-generated responses.
- Integrated Groq LLM APIs for context-aware response generation and Hugging Face FLUX.1-dev for text-to-image generation from user prompts.
- Implemented YouTube transcript-based summarization using LLMs to generate concise summaries from video content.
- **Tools Used:** Python, RAG, LLMs, Groq API, FAISS, Sentence Transformers, Hugging Face, FLUX.1-dev, PyPDF2, YouTube Transcript API

ML-Based Strabismus Risk Assessment System

- Developed a computer-vision-based system to identify potential eye misalignment from facial images using a CNN-based image classification approach.
- Integrated MobileNetV2-based transfer learning with a questionnaire-driven Logistic Regression model to combine image-based and clinical risk indicators.
- **Tools Used:** Python, TensorFlow/Keras, CNN, MobileNetV2, Transfer Learning, Logistic Regression, OpenCV, NumPy

Technical Skills

Programming Languages: Python, Java, C, SQL

AI / Machine Learning: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, Prompt Engineering, RAG, Semantic Embeddings, Vector Search

Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Sentence Transformers, FAISS

Tools & Technologies: FastAPI, REST APIs, Git, GitHub, VS Code

Certifications

- **Smart Shopping Cart Billing System Hackathon – TNWISE 2025**
- **Ethical Hacking Workshop – NIT Trichy**